

Angular Note

Install Angular CLI

```
npm install -g @angular/cli
```

Create new project

```
ng new my-site
```

Start the website

```
ng serve
```

Important Config Files

```
package.json  
tsconfig.json  
angular.json
```

Important Files

```
index.html  
main.ts
```

Creating New Component

```
ng generate component components/button
```

Add Fields in the component

```
import { Component, OnInit, Input } from '@angular/core';
```

```
@Component({  
  selector: 'app-button',
```

```

    templateUrl: './button.component.html',
    styleUrls: ['./button.component.css']
  })
  export class ButtonComponent implements OnInit {
    @Input() text:string;
    @Input() color:string;

    constructor() { }

    ngOnInit(): void {
    }
  }

```

In button component html, put this:

```

<input type="button" [ngStyle]="{'background-color':color}" value="{{text}}" (click)="onClick()" />

```

Add The Event Handler in button component.ts

```

onClick(){
    console.log("button clicked");
}

```

Use the button component in root component html:

```

<app-button text="Click-Me" color="red"></app-button>

```

Now Click the button to see output in console log

Now Let's add CSS Class as input:

```

<app-button text="Click-Me" cssClass="blueBorder" color="red"></app-button>

```

Put blueBorder in main styles.css

```

.blueBorder{
    border: 1px blue dashed;
}

```

In button component's html, put:

```
<input type="button" [ngClass]="cssClass" [ngStyle]="{'background-color':color}" value="{{text}}"
(click)="onClick()" />
```

Add another input in button component's class

```
@Input() cssClass:string;
```

Now We Convert This to EventEmitter

```
import { Component, OnInit, Input, Output, EventEmitter } from '@angular/core';
```

```
@Component({
  selector: 'app-button',
  templateUrl: './button.component.html',
  styleUrls: ['./button.component.css']
})
export class ButtonComponent implements OnInit {
  @Input() text:string;
  @Input() color:string;
  @Input() cssClass:string;
  @Output() btnClick = new EventEmitter();

  constructor() { }

  ngOnInit(): void {
  }

  onClick(){
    this.btnClick.emit();
  }
}
```

In App component we can now write:

```
<header>
  <h1>{{title}}</h1>
  <app-button cssClass="blueBorder" color="blue" text="Add"
(btnClick)="printMessage()"></app-button>
```

```
</header>
```

Add the method in App component's class

```
printMessenger() {  
    console.log("clicked");  
}
```

Now Click the button to see output in console log

Now Let's create a new component "Course"

ng generate component components/course

Add that component below our button

```
<header>  
    <h1>{{title}}</h1>  
    <app-button cssClass="blueBorder" text="Add" (btnClick)="printMessenger()"></app-button>  
    <app-course></app-course>  
</header>
```

Now Let us create a service

```
ng generate service services/course
```

Create ICourse interface and Course class in data folder:

```
export interface ICourse  
{  
    name : string;  
    fees : number;  
}
```

```
import { ICourse } from "./ICourse";  
  
export class Course implements ICourse{
```

```

    public name : string;
    public fees : number;

    // constructor syntax to easily initialize current object
    public constructor(init?:Partial<Course>) {
        Object.assign(this, init);
    }
}

```

Let's modify the service class to contain the data fetching logic

```

import { Injectable } from '@angular/core';
import { ICourse } from '../data/ICourse';
import { Course } from '../data/Course';

@Injectable({
  providedIn: 'root'
})
export class CourseService {

  constructor() { }

  getCourses() : ICourse[]{
    return [
      new Course({ name : "C#", fees : 8000 }),
      new Course({ name : "Asp.net", fees : 30000 })
    ];
  }
}

```

And modify the app component to use the service

```

import { Component } from '@angular/core';
import { ICourse } from '../data/ICourse';
import { CourseService } from '../services/course.service';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',

```

```

    styleUrls: ['./app.component.css']
  })
  export class AppComponent {
    title : string = 'first-demo-app';
    courses: ICourse[];

    constructor(private courseService: CourseService){ }

    update(){
      this.courses = this.courseService.getCourses();
    }
  }
}

```

Update App component html to add the course data in course component:

```

<h1>{{title}}</h1>
<app-button text="Click-Me" cssClass="blueBorder" color="red" (btnClick)="update()"></app-button>
<app-course [courses]="courses"></app-course>

```

Add following code in course component's html file:

```

<div *ngFor="let c of courses">
  {{c.name}} {{c.fees}}
</div>

```

Now click the button to see the output

Now let's add rxjs to make the method observable

Import rxjs in service

```

import { Observable, of } from 'rxjs';

```

Modify getCourses method

```

getCourses() : Observable<ICourse[]>{
  return of([

```

```

    new Course({ name : "C#", fees : 8000 }),
    new Course({ name : "Asp.net", fees : 30000 })
  ]);
}

```

Change update method

```

update(){
  this.courseService.getCourses().subscribe((data) =>
    (this.courses = data)
  );
}

```

Now click the button to see the course list through observable.

Connect With backend Server

Finally our service should look like:

```

import { Injectable } from '@angular/core';
import { ICourse } from '../data/ICourse';
import { Course } from '../data/Course';
import { Observable, of } from 'rxjs';
import { HttpClient } from '@angular/common/http';

@Injectable({
  providedIn: 'root'
})
export class CourseService {

  private tokenUrl : string = "/api/v3/token";
  private dataUrl: string = "/api/v3/course/getdata"
  private username: string = "api.user@gmail.com";
  private password: string = "Hello$DevSkill122";

  constructor(private http: HttpClient) { }

  getToken() : Observable<string>{
    let token = this.http.get(this.tokenUrl +

```

```

        `?email=${this.username}&password=${this.password}`,
        {
            headers: {'Accept': 'text/html, application/xhtml+xml,
*/*', 'Content-Type': 'application/x-www-form-urlencoded'},
            responseType: 'text'
        },
    );

    return token;
}

getCourses(token: string) : Observable<ICourse[]>{

    return this.http.get<ICourse[]>(this.dataUrl,
    {
        headers: {'Authorization': `Bearer ${token}`}
    },
    );
}
}

```

Update the app component's update method:

```

update() {
    this.courseService.getToken().subscribe((token) =>
    (
        this.courseService.getCourses(token).subscribe((data) =>
            (this.courses = data)
        )
    )
    );
}

```

Now Add the HttpClient Module in app.module file

```

import { HttpClientModule } from '@angular/common/http';

```


Also,

```
imports: [  
  BrowserModule,  
  AppRoutingModule,  
  HttpClientModule  
],
```

Now create a proxy file name : **"proxy.conf.json"** at the root of the site:

```
{  
  "/api/v3/*": {  
    "target": "https://localhost:7003",  
    "secure": false,  
    "logLevel": "debug",  
    "changeOrigin": true  
  }  
}
```

Then add the following line in **angular.json**

```
"serve": {  
  "builder": "@angular-devkit/build-angular:dev-server",  
  "options": {  
    "browserTarget": "firstdemo-front:build",  
    "proxyConfig": "proxy.conf.json"  
  },  
}
```

Make sure to add the url : <http://localhost:4200>

In the CORS rule of the API